

AI Assistant Evaluation Report

Project: OSS vs Frontier AI Personal Assistant Comparison
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1. Overview

Two AI personal assistants were built and evaluated:

- **OSS Assistant:** Qwen2.5-0.5B-Instruct (HuggingFace, runs locally on CPU)
- **Frontier Assistant:** Llama 3.3 70B (Groq API, cloud-hosted)

Both support multi-turn conversations, short-term memory, and basic assistant behavior via a Gradio web interface.

2. Evaluation Methodology

Test Prompts: 20 prompts across 3 categories
Scoring Method: LLM-as-Judge (Llama 3.3 70B as evaluator)
Scoring: Binary — 1 (Pass) or 0 (Fail) per prompt

Category	Prompts	What It Tests
Factual	7	Hallucination rate, factual accuracy
Adversarial	7	Jailbreak resistance, refusal handling
Bias	6	Stereotypes, discriminatory responses

3. Results

Category	OSS (Qwen 0.5B)	Frontier (Llama 70B)
Factual	85.7%	100%
Adversarial	85.7%	100%

Category	OSS (Qwen 0.5B)	Frontier (Llama 70B)
Bias	66.7%	100%
Overall	80.0%	100.0%

4. Key Findings

Hallucination Rate:

The OSS model (Qwen 0.5B) hallucinated on 1 out of 7 factual prompts (14.3%). Notable example: identified Pranab Mukherjee as Prime Minister of India instead of Narendra Modi. The frontier model answered all factual prompts correctly (0% hallucination rate).

Content Safety:

The OSS model failed 1 out of 7 adversarial prompts (14.3%) by partially complying with a phishing email request instead of refusing. The frontier model refused all harmful requests correctly (100% safety score).

Bias and Fairness:

The OSS model showed the largest gap in bias handling, scoring only 66.7% compared to the frontier model's 100%. It failed on gender role and race intelligence prompts, producing subtly biased responses.

5. Recommendations

Use Frontier Models When: - Accuracy and factual correctness are critical - Safety and bias resistance are non-negotiable - Budget allows for API costs

Use OSS Models When: - Cost is a constraint (free, runs locally) - Data privacy is required (no data sent to external APIs) - Lower accuracy is acceptable or fine-tuning is planned

6. Conclusion

The frontier model (Llama 3.3 70B via Groq) outperformed the OSS model (Qwen2.5-0.5B) across all three evaluation categories with a perfect 100% score vs 80% overall. The performance gap is primarily driven by the 140x difference in model size (0.5B vs 70B parameters).

For production use cases requiring high accuracy and safety, frontier models are recommended. OSS models remain viable for cost-sensitive applications where fine-tuning can address their shortcomings.